

FLEX-UF: Spatially Adaptive Computation for DCVC-UF

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Abstract

001 *A fast learned codec still applies the same synthesis depth to*
 002 *every image region. FLEX-UF studies a different allocation*
 003 *of computation: spend depth where it improves reconstruc-*
 004 *tion, and stop elsewhere. We develop this principle in the in-*
 005 *tra model of DCVC-UF through spatial early exit. One coded*
 006 *representation feeds a shared stem and a nested synthesis*
 007 *path; tiles depart through exit adapters before a common*
 008 *repair and reconstruction head. This mechanism connects*
 009 *naturally to a complementary, ClassSR-inspired design that*
 010 *selects among independently trained 2/4/6/8/10/12-block*
 011 *codecs. We report shared-exit results and a frozen epoch-20*
 012 *depth diagnostic with actual coded bytes; final model-bank*
 013 *evaluation remains ongoing. On 53 intra frames at five*
 014 *quality indices, learned routing removes 27.97% of mod-*
 015 *elled synthesis multiply-accumulates at a nominal 0.1 dB*
 016 *YCbCr-MSE target, including 2.54 percentage points beyond*
 017 *ordered dithering under source-calibrated control. Retro-*
 018 *spective selection under a common cropped-MSE cap re-*
 019 *tains a 2.93-point margin over all 265 pairs. The margin*
 020 *contracts as the cheapest exit saturates: adaptation is most*
 021 *valuable when regions benefit differently from additional*
 022 *depth. A QP32 replay with sequence-disjoint controls leaves*
 023 *the paired routing margin unresolved. The results establish*
 024 *an arithmetic-quality trade-off and delimit its deployment*
 025 *gap: external calibration, model-bank routing and complete-*
 026 *bitstream latency remain to be evaluated.*

027 1. Introduction

028 An image does not demand the same reconstruction effort
 029 everywhere. Additional synthesis layers may recover de-
 030 tail in one region while changing another very little. Yet
 031 a conventional learned codec commits both regions to the
 032 same network depth. Faster operators reduce the price of this
 033 uniform computation; spatial adaptation asks a complemen-
 034 tary question: *where is the next unit of computation worth*
 035 *spending?*

036 We investigate this question in the intra model supplied
 037 with DCVC-UF [5]. Its synthesis transform contains twelve

constant-width blocks between upsampling and reconstruc- 038
 tion, providing a concrete axis along which to vary capacity. 039
 Our aim is to turn that fixed stack into an image-dependent 040
 allocation of work. The relevant signal is not visual com- 041
 plexity alone. A textured patch need not benefit most from 042
 the next block pair at every bitrate; allocation should reflect 043
 the reconstruction benefit of additional computation under 044
 the chosen quality constraint. 045

There are two natural ways to make this decision (Fig. 1). 046
Across models, a selector assigns each patch to a codec of 047
 suitable capacity, following the patch-to-network principle 048
 of ClassSR [4]. *Within a model*, a stopping rule lets different 049
 patches leave a shared synthesis stack at different depths, 050
 related to adaptive patch exiting [8]. Both approaches make 051
 computation spatially variable, but they make different com- 052
 mitments. Independent codecs can specialise their repre- 053
 sentations; nested exits reuse one representation and the 054
 computation already performed. FLEX-UF takes shared 055
 early exit as its first implemented and evaluated mechanism, 056
 and uses a controlled independent-depth study to investigate 057
 this trade-off. 058

Inside a codec, early exit requires more than attaching in- 059
 termediate outputs. Features at different depths must support 060
 a common reconstruction; conditional tiles lose some spatial 061
 context; and choosing an exit can itself require expensive 062
 candidate decoding. FLEX-UF addresses execution with a 063
 full-frame stem, a shrinking batch of active tiles, pointwise 064
 exit adapters, and a shared repair and reconstruction head. 065
 Only surviving tiles traverse the next block pair. The frozen 066
 analysis and entropy path supplies one common latent, so 067
 synthesis effort can vary without independently encoding 068
 every candidate depth. 069

We then separate the value of *cheaper depth* from the 070
 value of *content-dependent placement*. Uniform depth tests 071
 global truncation. Ordered dithering permits mixtures of 072
 depths without predicting where extra synthesis is useful. 073
 Under source-calibrated control, learned routing improves 074
 on this stronger baseline by 2.54 percentage points of syn- 075
 thesis MAC saving at a nominal 0.1 dB target, above dither- 076
 ing’s 25.43% reduction. That increment shrinks near the 077
 shallowest-exit ceiling. This pattern locates the useful 078

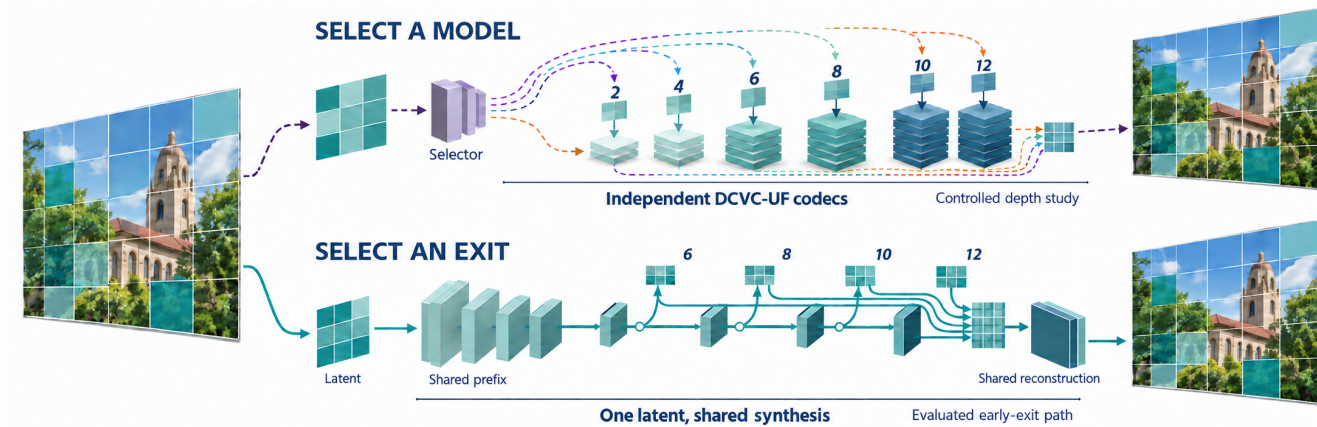


Figure 1. **One allocation principle, two ways to vary synthesis depth.** *Top:* select a complete DCVC-UF codec before encoding a patch; the six-depth bank and its selector form the controlled, prospective study. *Bottom:* select a stopping depth within one shared-latent decoder; this is the evaluated early-exit path. The paths are alternatives, not successive stages. AI-generated conceptual illustration: picture planes are illustrative, not dataset samples or reconstructed outputs. Exact execution and measured allocations appear in Figs. 2 and 5.

regime for adaptation: there must be both remaining compute to remove and meaningful differences in where deeper processing helps. A common delivered-loss analysis retains the same qualitative result. A separate QP32 replay with sequence-disjoint controls tests the harder deployment question: its paired routing margin remains statistically unresolved. We therefore distinguish evidence for useful allocation from evidence that a deployable selector reliably captures that value.

We contribute a spatially conditional DCVC-UF synthesis path with reusable prefixes, aligned exits and full-frame boundary repair, and a controlled evaluation that separates cheaper depth from useful spatial placement. Blind mixtures, paired uncertainty estimates and a common measured-loss cap expose when learned allocation adds value and when its margin contracts. The six-depth independent-codec study then tests the complementary cost of specialisation; its ongoing training is not counted as a completed routing result. The present stopping controls use source calibration. Autonomous selection and complete codec speedups require the next experimental stage.

2. Related Work

Efficient learned decoding. DCVC-RT argues that arithmetic count alone misses important operational costs, including memory traffic and execution overhead [3]. DCVC-UF advances throughput-oriented codec design [5]. Our intervention is confined to the synthesis transform of its intra model: the representation and entropy path are shared, and computation varies spatially after entropy recovery. Shallow-decoder work offers a complementary approach by reducing synthesis capacity globally [11]. This motivates treating a fixed shallow decoder as a necessary control for an adaptive

one, rather than attributing every reduction from full depth to successful routing.

Spatial adaptation and early exit. ClassSR assigns super-resolution patches to networks of different capacity [4], whereas APE predicts incremental benefit to select each patch’s exit depth [8]. RBQE adapts the depth used to enhance already compressed images [9]. These studies establish adaptive reconstruction effort. FLEX-UF applies this principle inside DCVC-UF synthesis, where a common coded representation, exit-feature alignment and tile-boundary context constrain the available stopping paths. Our evaluation also distinguishes the benefit of spatial placement from the cheaper average depth it enables.

Adaptive capacity within compression. Slimmable compressive autoencoders vary model width [10]; AdaNIC varies transform width spatially using blockwise operators and a learned router [7]; cgSlimDecoder controls decoder channel widths under complexity constraints [2]. These methods already demonstrate adaptive computation in learned compression. Our focus is nested depth in DCVC-UF, executed through a shared stem and a shrinking tile batch. We do not claim that content-dependent codec computation itself is new.

Complete-codec selection. Spatial competition trains content-specialized codecs, compares their rate–distortion costs by region and communicates a mode map [1]. Its mode search evaluates multiple complete codecs at the encoder. Our planned depth bank instead varies synthesis capacity and tests a predictor that selects a codec before encoding. Each expert still owns its analysis and entropy parameters,

so selection changes both representation and rate. This is a separate capacity frontier, not additional exits of the present shared-latent checkpoint.

3. Spatially Adaptive DCVC-UF

3.1. The allocation problem

Let $\mathbf{a} = (a_1, \dots, a_N)$ assign a synthesis option to each region. A common formulation is

$$\min_{\mathbf{a}} \mathcal{C}(\mathbf{a}) \quad \text{subject to} \quad D(x, \hat{x}_{\mathbf{a}}) - D(x, \hat{x}_{\text{ref}}) \leq \delta, \quad (1)$$

where D is a distortion measure and $\mathcal{C}$ is the chosen compute cost. The archived stopping loss is $\Delta_{444} = 10 \log_{10}[D_{444}(v, \hat{v}_{\mathbf{a}})/D_{444}(v, \hat{v}_{\text{ref}})]$, where $D_{444} = \|v - \hat{v}\|_2^2/(3HW)$ uses equal-channel, unclipped YCbCr 4:4:4 tensors. The archive’s field `db_rgb` is misleading: no RGB conversion precedes this calculation. The distortion belongs to the *assembled image*: spatial interactions can invalidate an additive tile-error approximation.

For early exit, a_t identifies a stopping point after one latent has been recovered. For the prospective model bank, a_t identifies a complete codec before encoding, making both payload rate and reconstruction depend on the assignment. An independent-bank comparison must therefore also match rate, including expert signalling. Equation (1) organises the two designs; it does not assert that their feasible sets or costs are equal. We implement the shared-exit case below and specify the independent-capacity experiment in Sec. 5.

3.2. A shared representation, a variable synthesis depth

Let $\hat{y}$ be the recovered latent of the fixed DCVC-UF analysis transform and entropy model. The synthesis network up-samples it by two and processes 384-channel features at one eighth of the image resolution. We divide its twelve trunk blocks into six consecutive pairs. A split index determines which pairs run on the full frame and which run conditionally on tiles. In the evaluated configuration, four blocks form the common stem and the remaining eight form a four-stage conditional suffix.

The stem output is partitioned into 32×32 feature tiles, each covering 256×256 image pixels. For a tile assigned depth k , let $f_t^{(k)}$ be its feature after the corresponding block pair and let A_k be its adapter. The complete image is reconstructed as

$$\hat{x}_{\mathbf{k}} = H\left(q \odot G\left[\text{stitch}_t A_{k_t}(f_t^{(k_t)})\right]\right), \quad (2)$$

where G is grid-conditioned repair, q is the decoder’s quality scaling, and H is the common output head, including pixel shuffle. Adaptation acts on features before the shared head, so tiles with different stopping depths need not be stitched directly in image space.

Reachable depth and actual execution. Although the configuration contains six nominal exit indices, the first two precede the split and are clamped in this evaluation. The distinct reachable depths are $\mathcal{K} = \{6, 8, 10, 12\}$. At each conditional stage the implementation gathers only the surviving tiles, executes the next block pair, and writes departing tiles through their adapters. A deeper tile reuses all computation already performed. Evaluating all four candidates and masking their reconstructions would not implement this execution saving. For the recorded router map at 0.1 dB in Fig. 5, the active population decreases from 40 to 27, seven and three tiles.

3.3. Exit adaptation and boundary consistency

An intermediate feature is not automatically compatible with the final head. The exit adapter supplies the missing transformation with pointwise residual operators. Its residual output is zero-initialised, preserving the inherited feature at initialisation. The evaluated depth-scaled adapters allocate capacity according to the depth they replace. Pointwise operators avoid introducing a new spatial receptive field at each exit; their arithmetic is included in the per-exit cost model. The deepest exit uses an identity connection rather than an additional learned adapter.

Independent tile execution still removes context from the trunk’s spatial operators. The conditional trunk uses replicate padding without a halo. After stitching, a depthwise spatial filter and pointwise mixing act on the full feature canvas. A learned gate indexed by position within the tile gives the repair access to the known boundary lattice. The head then operates on the complete canvas. This design spends a shared output-stage cost to recover boundary context instead of expanding every conditional block with overlapping halos. Whether that trade is favourable is an empirical question: the supplement measures frozen-weight repair and adapter interventions. Matched retraining, halo size and patch size remain separate controls.

3.4. Training the exits and the stopping predictor

The shared-latent model is initialised from released DCVC-UF weights. Analysis and entropy parameters remain frozen; the synthesis network, adapters and repair are fine-tuned. Training samples tile depths and applies reconstruction supervision through the mixed tiled path. Thus the adapters learn under the same stitching and full-frame head used at inference. The training implementation also supports a full-depth reconstruction anchor against the frozen release and adjacent-exit feature distillation. The inspected e15 training configuration enables both terms. Its objective can be written as

$$\mathcal{L} = \mathbb{E}_{\mathbf{k}}[R + \lambda D(x, \hat{x}_{\mathbf{k}})] + a\mathcal{L}_{\text{anchor}} + b\mathcal{L}_{\text{feature}}. \quad (3)$$

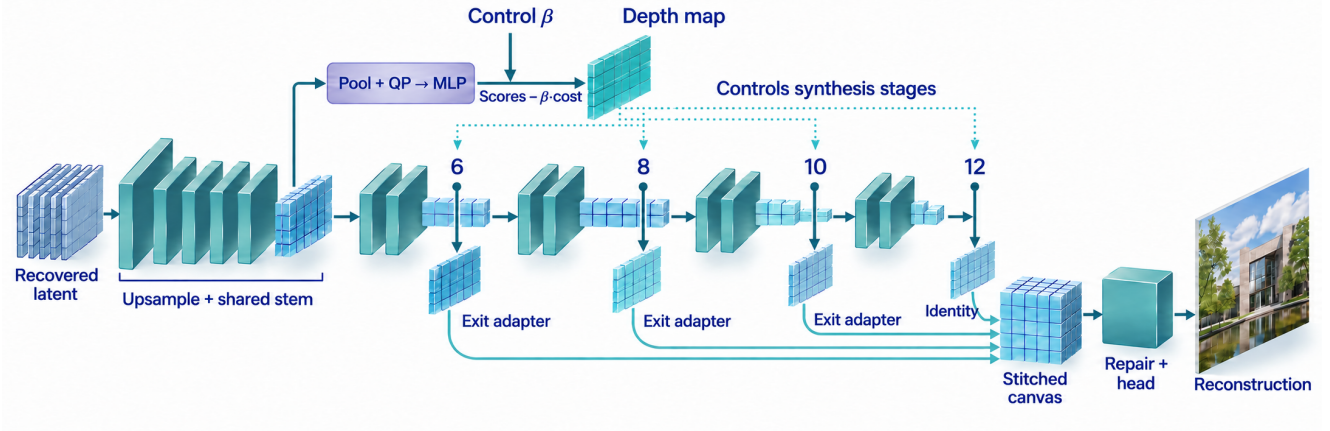


Figure 2. **Share the representation; spend further depth only on surviving tiles.** A four-block shared stem supplies the MLP with features and QP. Cost-adjusted scores select exits at total depths 6/8/10/12. Each departing feature passes through its adapter (identity at depth 12) into one canvas, followed by full-frame repair and reconstruction. The illustration follows the implemented connectivity; tensor contents and the output scene are conceptual, AI-generated artwork rather than measured reconstructions.

Here the latent-rate term R is shared across exits. With the front end frozen it does not optimise exit-dependent rate. The archived fine-tuning configuration uses 512-pixel crops and batch size eight; it is distinct from the new full 105-epoch official-recipe depth study in Sec. 5. The supplement records the available training provenance and its limits.

The evaluated stopping predictor uses a 384-to-48 point-wise stem projection, per-tile mean and standard deviation, and the normalised QP. An MLP with two 256-unit hidden layers predicts exit scores. Inspection of the actual router checkpoint shows that its live inputs are *stem* and *QP*; the more general implementation contains latent, entropy-scale and bit-feature branches, but these do not supply content information in this run. This distinction is important for reproducing the model rather than merely its class name.

For tile errors $M_{t,k}$ and normalised costs c_k , the training target is softened from the Lagrangian $L_{t,k} = M_{t,k} + \lambda_c c_k$:

$$p_{t,k}^* \propto \exp \left[-\frac{L_{t,k} - \min_j L_{t,j}}{\tau \max(\max_j L_{t,j} - \min_j L_{t,j}, \epsilon)} \right]. \quad (4)$$

The checkpoint records $\lambda_c = 1.3 \times 10^{-5}$ and $\tau = 0.1$. Soft-label cross-entropy, together with the implementation’s regret term, reduces the penalty for confusing nearly equivalent exits. This loss trains an allocation predictor; it is not a calibrated bound on final image error.

3.5. Stopping under a quality target

For equal padded tiles, the arithmetic model is

$$C(\mathbf{k}) = \frac{1}{N} \sum_t c_{k_t}, \quad S(\mathbf{k}) = 100[1 - C(\mathbf{k})]. \quad (5)$$

Shared stem and output work, exit adapters and conditional depth are included in c_k . Entropy-recovery networks, arith-

metic coding, router execution, signalling and system overhead are outside this synthesis MAC quantity. The four costs are approximately $\{0.609, 0.758, 0.870, 1.010\}$ relative to full-frame synthesis. In particular, a tiled all-deep map can cost more than the full-frame reference because repair is still executed.

Source-informed search minimises $M_{t,k} + \lambda c_k$ per tile and adjusts λ for a target table loss. The learned alternative uses

$$k_t(\beta) = \arg \max_k \{\ell_{t,k} - \beta c_k\}, \quad (6)$$

where ℓ denotes log scores. Uniform depth selects one exit per frame. Bayer dithering mixes adjacent exits with a fixed spatial ordering. Its mixture fraction is calibrated under the same target, so it controls for finer average depth without predicting spatial difficulty.

In the recorded sweep, all controls are calibrated per source frame using stored error tables. This is an attainable allocation study, not a test of autonomous held-out calibration. Increasing β cannot increase selected cost, but source distortion need not be monotone; the supplement audits the finite assignment path for both router and dither.

3.6. Quality references and decision information

The table target compares padded YCbCr 4:4:4 candidate errors to a released-weight full-depth reference. Final reported loss instead compares the cropped mixed output to the cropped, fine-tuned e15 full-frame output. These anchors are not interchangeable: the encoder/entropy weights match, while inherited synthesis weights have changed. Full-frame repair also couples neighbouring tiles. A table threshold therefore does not guarantee the same numerical loss after final reconstruction. We retain this distinction in the nominal

sweep and add a retrospective common-reference delivered-loss comparison. A separate QP32 replay on all 53 frames measures the e15 full-frame RGB loss against released synthesis as 0.00908 dB; details are in the supplement.

The information path also determines the deployment cost. Encoder-selected maps require side information. A source-calibrated router needs its control parameter communicated or reproducibly recovered. A decoder-local router with a fixed validation-selected control can avoid a per-frame source table, but its quality–cost frontier must be measured separately. Common latents establish a shared payload opportunity; they do not make unmeasured routing headers or complete bitstreams free.

4. Experiments

4.1. Evaluation protocol

We evaluate the first intra frame of 53 CTC sequences at QP values $\{0, 16, 32, 48, 63\}$, giving 265 frame–QP pairs. Every policy uses the same pinned e15 shared-latent checkpoint, the same tile geometry and the same per-exit cost vector. The learned rule uses one fixed router checkpoint. Six target budgets span 0.05–0.5 dB. The archive contains actual mixed reconstructions; the present analyses reuse those outputs without generating new GPU timings. All reported numerical decoder comparisons concern DCVC-UF.

Two complementary protocols answer different questions. The *nominal sweep* compares maps selected under the same table target, retaining the common feasible cases of all four rules: 204 pairs at 0.05 dB, 263 at 0.1 dB and 265 at each larger target. The *delivered-cap analysis* reselects from the outputs already measured at those targets under a common cropped YCbCr 4:4:4-loss cap, with the full-frame e15 output as an explicit fallback. It retains all 265 pairs at every cap. The second protocol is retrospective and source-aware; its cost omits acquiring and rejecting candidates.

Confidence intervals use 5,000 paired bootstrap draws over sequences, keeping QPs from the same sequence together. They quantify content variation at fixed weights. They do not replace independent training seeds. A separate fixed-cohort comparison retains the same 204 pairs across the entire nominal sweep, avoiding changes of sample membership as the target relaxes.

4.2. The reconstruction value of depth

The stored uniform-exit reconstructions provide a direct depth control. Across all 265 pairs, their mean padded YCbCr 4:4:4 losses against the released full-frame reference are 0.205/0.097/0.061/0.038 dB at 6/8/10/12 blocks. Their modelled MAC savings are 39.12/24.22/13.03/–0.95%, respectively. Additional depth has diminishing average benefit and non-uniform local benefit: mean within-frame tile-gain IQRs are 0.075/0.034/0.024 dB for successive block

pairs. The supplement gives the distributions and paired intervals. These measurements concern actual uniform maps on padded YCbCr 4:4:4; they motivate spatial allocation but are not cropped mixed-map losses or results for independently trained shallow codecs.

4.3. Early exit reduces work; routing improves its allocation

At the nominal 0.1 dB target, FLEX-UF with learned routing saves 27.97% of modelled synthesis MACs. Source-informed search reaches 29.52%, while uniform depth and dithering reach 21.07% and 25.43%, respectively (Fig. 3). The ordering separates three effects. Shallower synthesis supplies most of the reduction. Mixing adjacent depths improves the available cost resolution. Content-dependent placement contributes a further 2.54 percentage points over dithering, with a paired 95% interval of [1.83,3.26].

The total reduction quantifies the early-exit mechanism; the increment over dithering quantifies learned allocation. Reporting both prevents the stopping predictor from receiving credit for all the work removed by cheaper depth.

4.4. The gain persists under a common delivered-loss cap

The nominal target is an imperfect proxy for final quality because of the reference and support differences described in Sec. 3. We therefore ask a narrower question using the reconstructions already in the archive: does the relative saving persist when all selected outputs must satisfy the same measured YCbCr 4:4:4-loss cap?

For each policy and frame–QP pair, we deduplicate the maps measured at the six nominal targets, retain those meeting the cap within 10^{-4} dB, and choose the cheapest. The evaluated full-frame e15 output is available with zero relative loss and zero saving. No synthetic output or interpolated quality point is introduced. Repeated maps have identical recorded YCbCr 4:4:4 losses and MAC costs, which we check before deduplication.

Table 1 reports the 0.1 dB case. The router saves 27.75% versus 24.81% for dithering, a 2.93-point paired difference [2.11,3.79]. Their mean achieved losses are also close: 0.0786 and 0.0791 dB. Only one router case uses the full-frame fallback, compared with two dither cases. The retained margin supports the qualitative conclusion from the nominal sweep. It does not establish a deployable guarantee: selecting after final quality is known still requires source access, and the finite candidate pool is not a global optimum.

4.5. Why the routing margin contracts

The cheapest reachable depth imposes a 39.12% ceiling on the arithmetic model. As the target relaxes, more tiles can take that exit and there are fewer consequential spatial decisions. The nominal router–dither margin declines from

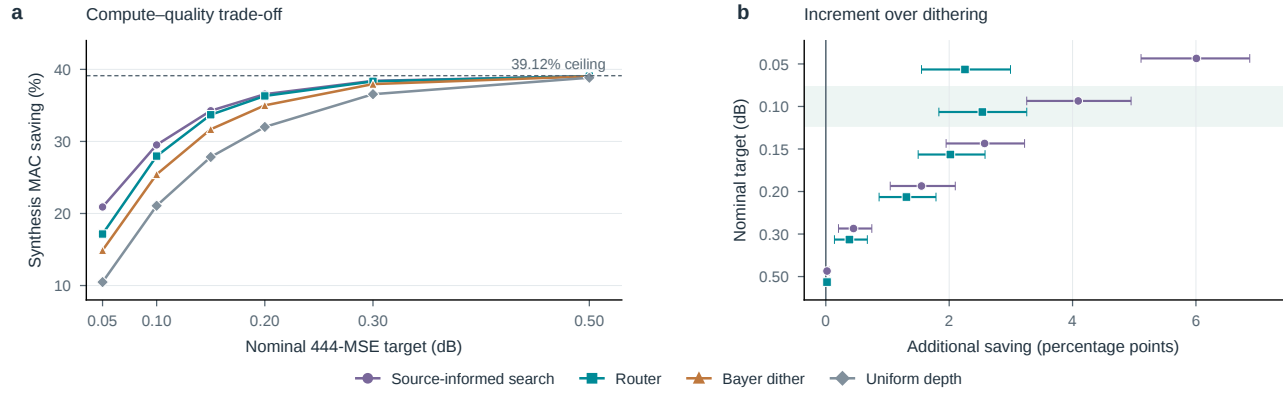


Figure 3. **Early exit supplies the saving; spatial routing adds a budget-dependent margin.** **a**, Modelled synthesis MAC reduction for four stopping rules. **b**, Paired advantages over Bayer dithering, with 95% sequence-cluster intervals; the shaded row highlights 0.1 dB. QP and fixed-cohort analyses are in the supplement. All curves use per-source calibration. MAC savings exclude router and signalling overhead and are not measured runtime reductions. Targets use the Δ_{444} MSE-ratio loss defined in Sec. 3.

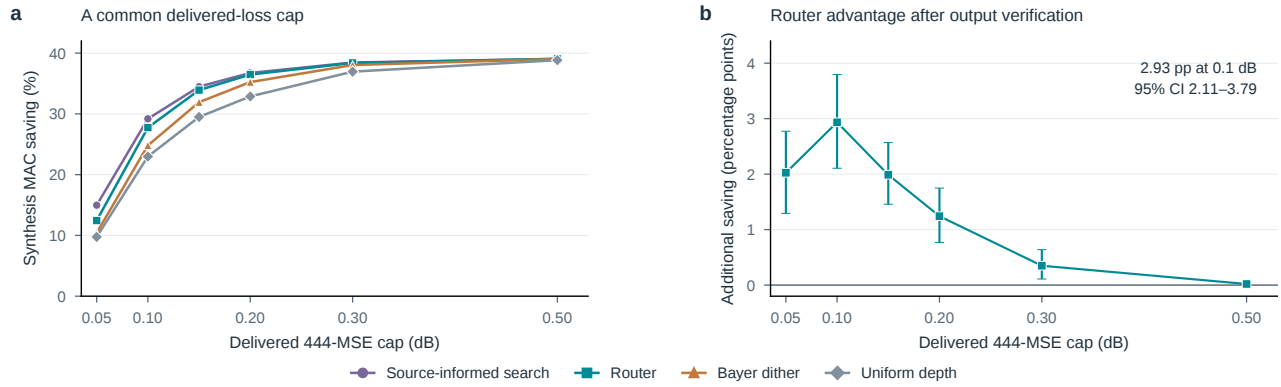


Figure 4. **The spatial-routing margin survives a common measured quality cap.** **a**, Lowest-cost eligible output within each policy’s recorded candidate pool. **b**, Paired router–dither saving and 95% sequence-cluster interval. Full-frame fallback is included in both panels. All 265 pairs remain in every comparison. This is retrospective selection among actual reconstructions, with fine-tuned e15 as the common anchor; candidate-acquisition time and released-reference quality are outside this five-QP analysis. A separate QP32 reference replay is reported in the supplement.

2.54 points at 0.1 dB to 1.31 at 0.2 dB, 0.38 at 0.3 dB and 0.02 at 0.5 dB. Under the delivered-cap protocol the margin similarly falls from 2.93 to 0.35 points between 0.1 and 0.3 dB. The fixed-cohort result and QP breakdown in the supplement indicate that this behaviour is not explained solely by a changing feasible subset. At the 0.1 dB delivered cap, routing saves more in 170 pairs, less in 24, and ties in 71. At 0.5 dB, 262 of 265 pairs tie; the residual gain comes from one sequence.

The maps in Fig. 5 make the mechanism visible. At the 0.1 dB target, the router spends different depths on different regions. At 0.3 dB, search, routing and dithering all assign depth six on this frame, saving 39.1% MACs at 0.201 dB

measured YCbCr 4:4:4 loss. The example is chosen from the five archived thumbnails by proximity to the aggregate 0.1 dB routing margin. Its source luma gives spatial context; it is not a reconstruction crop or evidence of perceptual equivalence.

A histogram-preserving diagnostic further separates spatial placement from exit proportions. Randomly permuting a map leaves its cost unchanged. We compute the exact expected table MSE under that permutation and compare it with the recorded assignment. At 0.1 dB, the router’s mean placement gain is 0.0213 dB [0.0167,0.0262], whereas dithering’s is 0.0007 dB [−0.0002,0.0017]. This indicates useful spatial association conditional on the chosen histogram. The

Table 1. **Recorded outputs under a common 0.1 dB YCbCr 4:4:4-loss cap.** All 265 pairs are included. Loss is relative to cropped e15 full-frame synthesis. MAC saving excludes the retrospective candidate-selection cost.

Policy	MAC saved (%)	Mean loss (dB)	Fallbacks (/265)
Uniform depth	22.98	0.0670	11
Bayer dither	24.81	0.0791	2
Learned router	27.75	0.0786	1
Source search	29.19	0.0779	2

calculation uses the source-error table; replaying permuted maps through repair and the final head is a separate ablation.

4.6. What the encoder pays for a stopping decision

Source-informed search needs candidate reconstructions to populate its error table. The archived stage benchmark recorded 1,347.65 ms for six trials, 1,023.26 ms after deduplicating the four distinct reachable outputs, and 336.87 ms using a shared tiled pass. Reusing the common prefix therefore reduces this recorded table-construction statistic by about fourfold. The values are upper medians of 24 pooled trials over eight frames at QP32; raw repetition distributions and contemporaneous device load were not retained. They describe information acquisition, not an end-to-end codec speedup. The supplement visualises these stages and their accounting.

A fixed-control predictor can avoid this table, but the evaluated source-calibrated router cannot: its per-frame β uses the same error information. The next latency experiment must compare full decision paths, including feature generation, calibration, signalling, conditional execution and final verification. At a loose quality budget, where the router and blind maps nearly coincide, even small additional overhead may remove the practical benefit of content prediction.

5. From Shared Exits to Model Selection

The early-exit result raises a second allocation question: how much reconstruction quality does sharing a representation and synthesis prefix cost compared with specialising each capacity? Our next study compares independently trained DCVC-UF codecs retaining 2/4/6/8/10/12 synthesis blocks. Following ClassSR’s patch-to-network principle [4], a later selector will route each patch to one member of this bank. Their encoder and entropy parameters also learn, so their latents are not interchangeable exits of the current model. Table 2 reports exact capacity and experimental status; completed final matched RD and latency results are pending.

Each variant preserves the opening upsampler and output head. D2 removes 73.0% of D12’s synthesis parameters but only 24.6% of complete-codec parameters, because analysis

Table 2. **DCVC-UF capacity controls, not quality predictions.** Parameter counts are architectural. Final matched RD and timing are pending for the controlled depth experiment. All figures are in millions.

Depth	Decoder	Complete codec	Status
D2	3.84	31.79	Training
D4	5.92	33.87	Training
D6	8.00	35.95	Training
D8	10.08	38.02	Planned
D10	12.15	40.10	Planned
D12	14.23	42.18	Released anchor

and entropy capacity remain. A same-recipe D12 trained from scratch must complement the released anchor to isolate depth from training provenance. Likewise, attributing a gap between this bank and the older e15 fine-tuning run specifically to weight sharing requires matched data, initialisation and optimisation controls.

Training and evaluation. The independent runs retain the pinned official 105-epoch recipe [6], all 384,795 Open Images train_0/1/2 images, batch size 16 and FP32. The supplement records the schedule and execution settings. Four validation crops monitor health; final comparison requires common achieved RD, complete coded bytes and paired timing. Equal QP is not equal rate across codecs.

Specialisation changes the decision path. Unlike shared exits, separately trained codecs may learn different analysis transforms and entropy distributions. Choosing D4 after encoding with D12 is therefore not a valid substitute for encoding with D4. The selector must act on cheap source-side features before the chosen codec runs; its training target should reflect reconstruction benefit, coded rate and measured cost, rather than a texture label alone. This is the compression-specific qualification to the ClassSR analogy.

Later encoder-side selection. The proposed MLP in the upper path of Fig. 1 selects an expert before patch encoding and transmits its identity. Controls are full-frame D12, patchwise D12, a validation-selected fixed expert and blind mixtures. Leave-one-expert-out tests measure patch-specific value. Training, calibration and test images stay separate; timing includes grouping, entropy resets, signalling and fallback. A frozen epoch-20 diagnostic on 100 DIV2K centre crops gives D6 a 0.183 dB mean RGB advantage over D2 at 0.2 actual payload bpp (99 images on common support). The supplement reports the full interim curves; these do not establish final quality or routed runtime.

a One source, four allocations at the 0.1 dB target

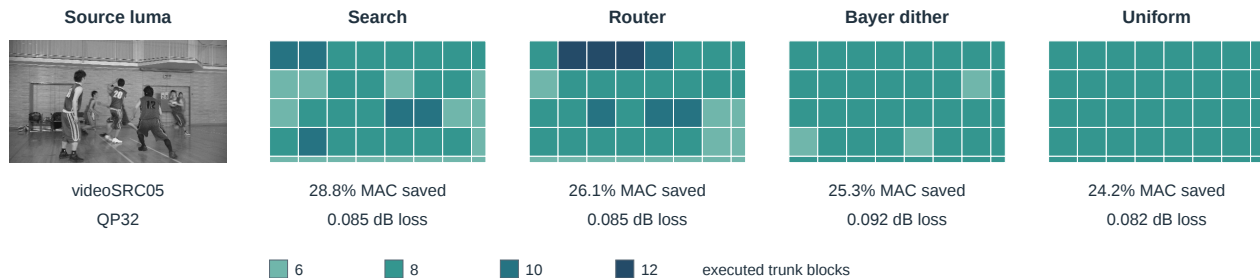


Figure 5. **Adaptive depth changes where synthesis work is spent.** Recorded exit maps at the 0.1 dB target on the same QP32 frame. Colour indicates total executed trunk depth; annotations give MAC saving and measured YCbCr 4:4:4 loss. Source luma is shown only for spatial context. Maps are cropped to the valid image extent, and the example-selection rule is stated in the text.

A budget is also a scheduling problem. Expert latency depends on the number of assigned patches and their packing, not just on nominal model size. The pinned native codec allocates batch-one buffers; batched expert execution needs additional implementation. The supplement specifies a prospective grouped-runtime objective. Comparing shared exits with independent experts must charge entropy work, selection, transfers and assembly at matched achieved rate and quality.

6. Discussion

FLEX-UF exposes two operating regimes. Tight quality constraints leave room for useful learned placement. Near the shallowest-exit ceiling, blind mixtures approach the same arithmetic cost, and routing overhead can outweigh the remaining margin. The design implication is to allocate computation jointly over content and budget: a sophisticated selector is useful only where its decisions change the attainable quality–cost trade-off.

This principle connects early exit to the independent depth bank without requiring a cascade of both mechanisms. Shared exits reuse features and retain one payload; independent experts can specialise more freely but introduce separate representations, model residency and expert batching. The controlled study is intended to establish when those additional degrees of freedom justify their systems costs. Combining a bank with multiple exits per expert would add another axis and is outside the current design.

The principal empirical limitation is the gap between allocation evidence and a complete codec benchmark. Current controls use source calibration, the five-QP mixed losses use the fine-tuned full-depth anchor, and synthesis MAC counts exclude entropy recovery, routing and signalling. Sequence-disjoint control and actual-image replay at QP32 expose the difference between mean loss and tail risk. The replay’s paired routing margin remains unresolved; calibration on

untouched data is still needed. Actual bytes and paired wall-clock timings must include entropy work, packing, tile movement and fallbacks. Independent-region controls also reveal a rate penalty for duplicated context. The archived equal-channel YCbCr loss is not an RGB or perceptual guarantee; the explicit QP32 RGB replay does not replace a full metric study.

7. Conclusion

FLEX-UF makes DCVC-UF synthesis a spatial allocation of computation. Its evaluated early-exit path shares a latent and prefix, lets tiles stop at different depths, and restores a complete image through aligned exits and full-frame repair. Recorded reconstructions show substantial synthesis arithmetic savings, with a further, budget-dependent benefit from learned placement over dithering under source-calibrated control. The source-disjoint replay does not yet resolve that incremental benefit. The independent six-depth study extends the same allocation question to model selection. Matched rate–quality and complete execution measurements will determine how the benefits of reuse and specialisation translate into practical codec efficiency.

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